

WEEK – 3

Java programming -Java Buzzwords, Data types,
Variables, Operators, Control structures,
Arrays, Console input and output

Sample Program:

1. Sample Java program to check whether the given number is prime palindrome or not. CO1.
2. Write a java program which reads marks of five subjects through command line and print the total and average.

Steps of the Code

Step 1: Import Scanner

```
import java.util.Scanner;
```

This allows the program to take input from the keyboard.

Step 2: Create primepalindrome class

This class contains two methods:

isPrime() → checks whether a number is prime.

isPalindrome() → checks whether a number reads the same forward and backward.

Step 3: Check Prime Number

The isPrime(int n) method:

If $n \leq 1$, return false.

If $n \leq 3$, return true.

Check whether the number is divisible by 2 or 3.

Use a loop to check possible factors.

If a factor is found, return false.

Otherwise return true.

Step 4: Check Palindrome

The isPalindrome(int n) method:

Store the original number.

Set reversed = 0.

Extract the last digit using:

```
digit = n % 10;
```

Add the digit to the reversed number.

Remove the last digit using:

```
n /= 10;
```

Repeat until n becomes 0.

Compare original with reversed.

If equal, the number is a palindrome.

Step 5: Take Input

```
Scanner pp = new Scanner(System.in);
```

```
int n = pp.nextInt();
```

The user enters a number.

Step 6: Call Both Methods

```
boolean prime = primepalindrome.isPrime(n);
```

```
boolean palindrome =
```

```
primepalindrome.isPalindrome(n);
```

Step 7: Display Result

There are four possibilities:

Prime

Palindrome

Result

Yes

Yes

Prime and Palindrome

Yes

No

Prime but not Palindrome

No

Yes

Not Prime but Palindrome

No

No

Neither

Step 8: Close Scanner

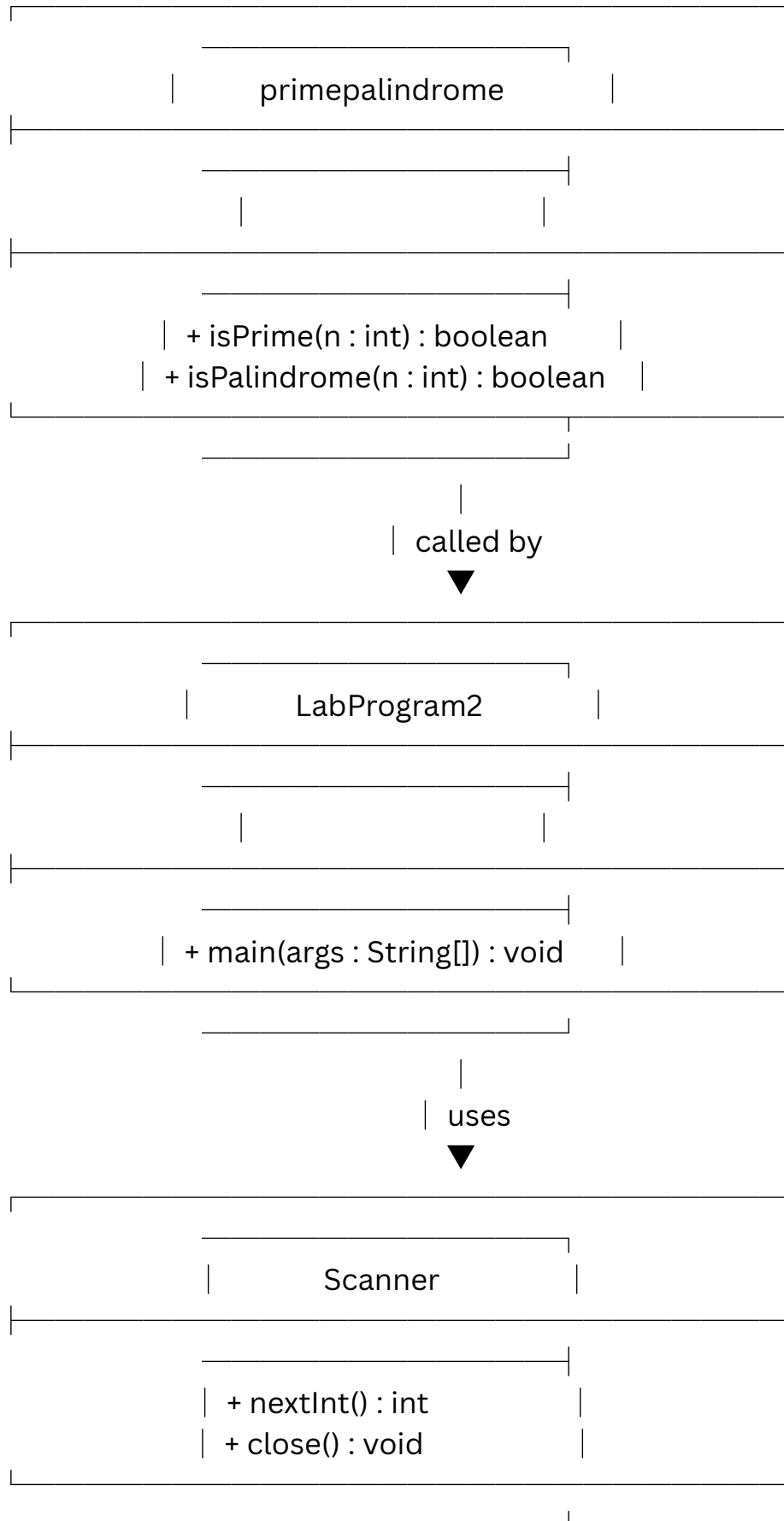
```
pp.close();
```

START

1. Read the number n.
2. Call isPrime(n).
3. In isPrime(n):
 - a. If $n \leq 1$, return false.
 - b. If $n \leq 3$, return true.
 - c. If n is divisible by 2 or 3, return false.
 - d. Check divisibility using a loop.
 - e. If a divisor is found, return false.
 - f. Otherwise return true.
4. Call isPalindrome(n).
5. In isPalindrome(n):
 - a. Store n in original.
 - b. Set reversed = 0.
 - c. Repeat while $n \neq 0$:
 - i. $digit = n \% 10$
 - ii. $reversed = reversed * 10 + digit$
 - iii. $n = n / 10$
 - d. Compare original and reversed.
 - e. If equal, return true.
 - f. Otherwise return false.
6. If prime == true AND palindrome == true:
Print "PRIME NUMBER AND PALINDROME".
7. Else if prime == true:
Print "PRIME BUT NOT PALINDROME".
8. Else if palindrome == true:
Print "NOT PRIME BUT PALINDROME".
9. Otherwise:
Print "NEITHER PRIME NOR PALINDROME".
10. Close Scanner.

STOP

UML DIAGRAMS



```

import java.util.Scanner;

class primepalindrome{
static boolean isPrime(int n){ // method to check
    prime number
    if(n <= 1)
        return false;
    if(n<= 3)
        return true;
    if(n %2 == 0 || n%3 == 0)
        return false;
    for (int i = 5 ;(long) i * i<= n;i +=6){
        if(n % 2 == 0 || n %(i +2) == 0)
            return false;
        }
    return true;
    }

static boolean isPalindrome(int n){
    int original = n;
    int reversed = 0;
    while(n != 0){
        int digit = n%10;
        reversed = reversed * 10+ digit;
        n /= 10;
    }
    return original == reversed;
    }
}

public class LabProgram2 {
    public static void main(String[] args) {
        Scanner pp = new Scanner(System.in);
        System.out.println("enter the number");
        int n = pp.nextInt();
        boolean prime = primepalindrome.isPrime(n);
        boolean palindrome =
            primepalindrome.isPalindrome(n);
        if(prime && palindrome){
            System.out.println( n +" is a PRIME NUMBER AND
                PALINDROME");
        }
        else if(prime){
            System.out.println(n + "IT IS PRIME BUT NOT
                PALINDROME");
        }else if(palindrome){
            System.out.println(n + "IS NOT PRIME BUT IT IS A
                PALINDROME");
        }else{
            System.out.println(n + "IS NEITHER PRIME NOR BE
                PALINDROME");
        }
        pp.close();
    }
}

```

Output of code

enter the number

121

121 is a PRIME NUMBER AND PALINDROME

[Program finished]

. Steps of the Program

Step 1: Start the program

Execution begins from:

```
public static void main(String[] args)
```

Step 2: Check command-line arguments

The program checks whether exactly 5 marks are entered.

```
if (args.length != 5)
```

If the number of arguments is not 5, it displays the usage message and stops.

Step 3: Create variables

```
double total = 0;
```

```
double[] marks = new double[5];
```

total stores the sum of all marks.

marks[] stores the five subject marks.

Step 4: Read the five marks

The for loop runs from 0 to 4.

```
for (int i = 0; i < 5; i++)
```

Each command-line argument is converted from String to double using:

```
Double.parseDouble(args[i])
```

Step 5: Calculate total

Each mark is added to total.

```
total += marks[i];
```

Step 6: Calculate average

```
average = total / 5;
```

Step 7: Display marks

A second for loop displays each subject's mark.

Step 8: Display total and average

Finally, the program displays:

Total marks

Average marks

Step 9: Stop

The program finishes execution.

START

1. Read five marks from command-line arguments.
2. Check whether the number of arguments is equal to 5.
 3. If the number of arguments is not 5:
Display usage message.
Stop the program.
4. Initialize:
total = 0
5. Create an array marks[5].
6. Repeat for i = 0 to 4:
 - a. Convert args[i] from String to double.
 - b. Store it in marks[i].
 - c. Add marks[i] to total.
7. Calculate:
average = total / 5
8. Display "Marks entered".
9. Repeat for i = 0 to 4:
Display Subject i+1 and marks[i].
10. Display total.
11. Display average.
12. STOP



```
public class LabProgram21 {  
    public static void main(String[] args) {  
        if(args.length != 5){  
System.out.println("usage: java marks <m1> <m2>  
        <m3> <m4> <m5>");  
            // return;  
        }  
        double total = 0;  
        double[] marks = new double[5];  
        for (int i = 0 ; i < 5; i++){  
marks[i] = Double.parseDouble(args[i]);  
            total += marks[i];  
        }  
        double average = total /5;  
  
System.out.println("marks entered");  
  
        for(int i = 0 ; i < 5 ;i++){  
System.out.println("subject" + (i+1) + ":" + marks[i]);  
        }  
        System.out.println(" total is : " +total);  
System.out.println("average is :" + average);  
        }  
    }
```

Output of code

```
javac LabProgram21.java
```

```
java LabProgram21 85 90 78 92 88
```

```
Marks entered  
Subject 1: 85.0  
Subject 2: 90.0  
Subject 3: 78.0  
Subject 4: 92.0  
Subject 5: 88.0  
Total is: 433.0  
Average is: 86.6
```